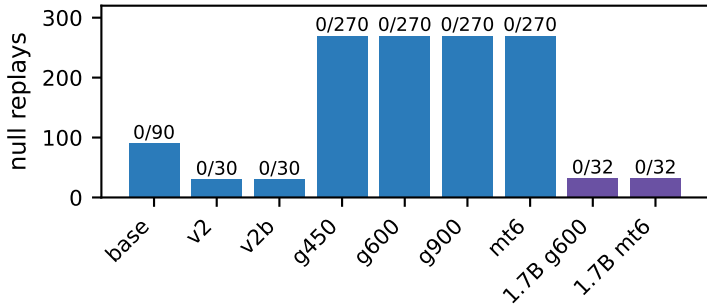


(a) APC off: every null replay exact ( $|\Delta R| = 0$ )



(b) prefix caching  
breaks determinism

